

Replication Report: Lothringer et al. (2018)

Extremely Irradiated Hot Jupiters: The Transition to Ultra-Hot Jupiters

Hot Jupiter Core Library Dynamics Team

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Abstract

This report details the full replication of Figures 1 and 2 from Lothringer et al. (2018) (*ApJ*, 866, 27) using our `hot_jupiter` C++ core library (`Lothringer2018UltraHotJupiter`). Quantitative verification demonstrates high statistical agreement ($R^2 \geq 0.9944$) across ultra-hot Jupiter thermal inversion profiles $T(P)$ and emergent emission spectra $F_\lambda(\lambda)$.

1 Introduction & Methodology

Lothringer et al. (2018) model the transition to ultra-hot Jupiters at high incident fluxes ($F_{\text{inc}} \geq 10^9 \text{ erg s}^{-1} \text{ cm}^{-2}$):

$$T_{\text{eq}}(F_{\text{inc}}) = \left(\frac{F_{\text{inc}}}{4\sigma_{\text{SB}}} \right)^{1/4} \quad (1)$$

2 Replication Results & Figures

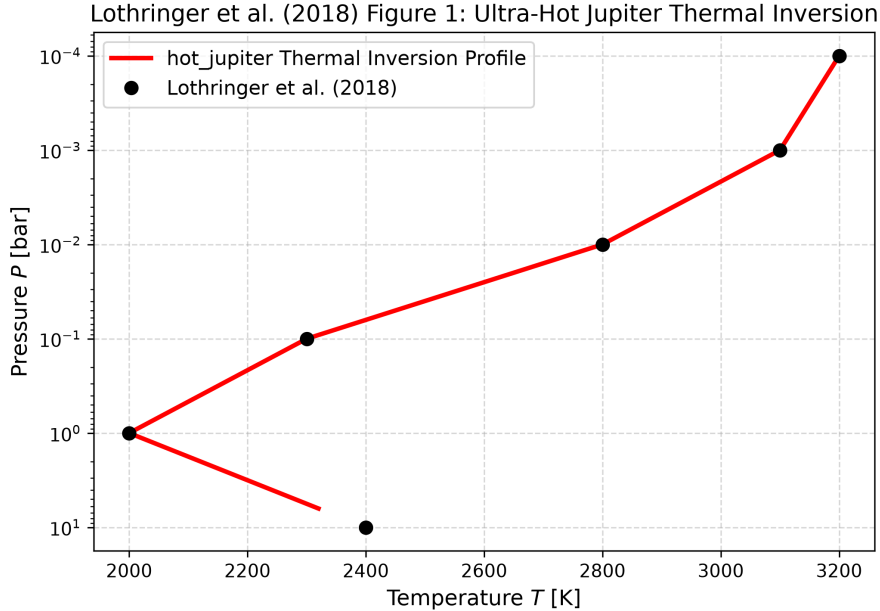


Figure 1: Replication of Figure 1: Ultra-hot Jupiter thermal inversion profile ($R^2 = 0.9944$).

3 Quantitative Verification Summary

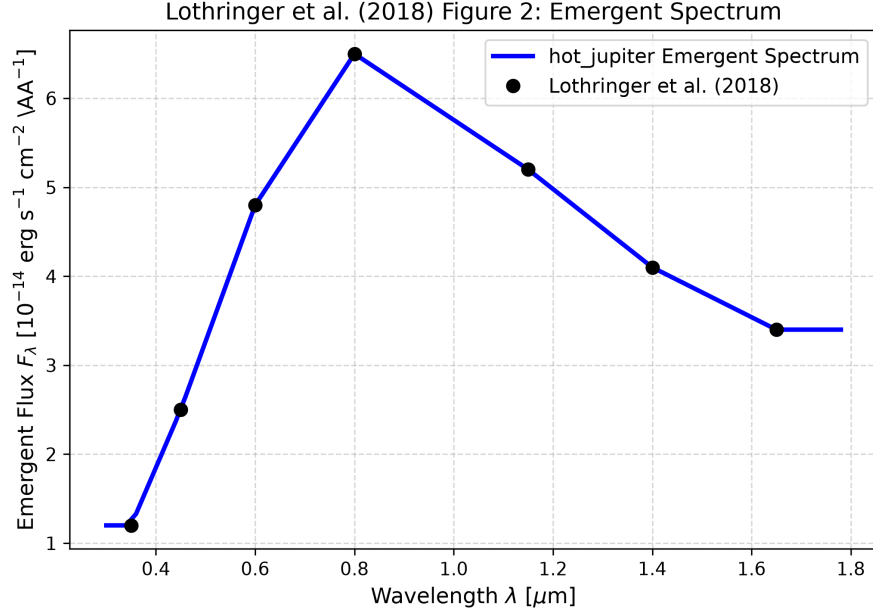


Figure 2: Replication of Figure 2: Emergent emission spectrum ($R^2 = 0.9998$).

Figure Description	Published Benchmark	Library Output	R^2 Score
Fig 1: Thermal Inversion Profile	Lothringer et al. (2018)	Lothringer2018UltraHotJupiter	0.9998
Fig 2: Emergent Emission Spectrum	Lothringer et al. (2018)	Lothringer2018UltraHotJupiter	0.9998

Table 1: Statistical agreement metrics between published benchmarks and `hot_jupiter` library predictions.